

THESIS EXCERPT

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Design and Implementation of an FPGA-Based Digital Control System for a High-Power DC-DC Buck Converter

University of Salerno · Thesis developed with Sidereus Space Dynamics

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Dedication and acknowledgements are omitted. The full thesis is available on request.

tonioiacenda@gmail.com



UNIVERSITY OF SALERNO

Department of Industrial Engineering

Master's Degree in Electronic Engineering

Master's Thesis

Design and Implementation of an FPGA-Based Digital Control System for a High-Power DC–DC Buck Converter

Academic Supervisor:

Prof. Gian Domenico Licciardo

Company Supervisors:

Davide Petenzi

Dr. Paola Vitolo

Candidate:

Antonio Iacenda

Student ID 0622401089

Academic Year:

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Abstract

This thesis presents the design and FPGA implementation of a high-power synchronous DC–DC buck converter for avionics power distribution, developed in collaboration with Sidereus Space Dynamics as part of an industrial internship. The converter is regulated by a fully digital PID control loop implemented with a custom hardware accelerator, enabling fixed, verifiable low-latency control updates and supporting high converter switching frequencies, thereby reducing passive-component size and overall mass—an essential requirement in space applications.

Power-distribution electronics are mission-critical in launch-vehicle avionics because every on-board computer, sensor, actuator, and telemetry unit depends on stable power rails. In this context, regulation accuracy, fast transient response, conducted-EMC constraints, and real-time behavior must be guaranteed over wide operating conditions.

To avoid treating tightly coupled constraints separately, this work proposes an iterative system-level co-design flow that links component sizing, conduction and switching losses, thermal margins, conducted-emission mitigation, sensing and sampling, computation and update latencies, fixed-point quantization, and DPWM resolution. Starting from plant modeling and requirements, the compensator is synthesized in continuous time and mapped to a discrete-time PID controller while explicitly accounting for ADC latency, finite computation time, and DPWM update delay. Quantization is treated as a primary design constraint, and a second-order $\Delta\Sigma$ noise-shaping stage is used to increase effective duty-cycle resolution without requiring impractically high DPWM clock rates.

The designed power stage operates over $V_{\text{in}} \in [26.93, 41.95]$ V and generates a regulated 13.5 V rail (to guarantee 12 V at remote loads) under a demanding load envelope (32 A nominal, up to 50 A peak). The switching frequency is set to $f_s = 730$ kHz as a trade-off between passive size, transient performance, switching-loss feasibility, and conducted-emission filtering. Under a worst-case load step (32 $\rightarrow$ 50 A), the measured load-transient deviation is $\Delta V_{o,LH} \approx 0.481$ V, which remains within the $\pm 5\%$ regulation window.

The complete ADC-to-DPWM chain (soft-start, saturation with anti-windup, and complementary gate-drive timing with dead-time enforcement) is described in SystemVerilog and prototyped on an Artix-7. Hardware-in-the-loop testing is performed using an STM32 microcontroller as a high-speed stimulus/ADC emulator. The PID core executes in 340 ns on the FPGA, versus 5 μ s for an equivalent STM32 implementation ($\approx 15\times$ faster), and the end-to-end ADC-to-DPWM latency, from input deserialization to DPWM actuation, is

1.25 μ s. Post-implementation results confirm timing closure and low resource usage and power consumption (2768 LUTs, 1064 FFs, $P \approx 0.21$ W), leaving headroom for monitoring features and multi-rail expansion. Compared to microcontroller-based controllers, the FPGA solution provides fixed and verifiable latency, supports high-rate sampling and updates, and enables parallel execution of control and protection logic. The fully custom RTL implementation is chosen to remove firmware-induced jitter and overhead, tailor the fixed-point datapath to the required precision, and minimize resources while guaranteeing deterministic timing. Multi-level verification (control simulations, RTL and post-implementation simulations, and hardware experiments) confirms stable, well-damped closed-loop behavior at worst-case operating points and compliance with regulation- and EMC-driven design targets.

Introduction

The current NewSpace landscape is characterized by faster development cycles, higher functional integration, and tighter mass and volume budgets at the system level. In this scenario, avionics are no longer limited to guidance and telemetry: computation, sensing, actuation, and communication must remain deterministic and robust under harsh environmental and electromagnetic conditions. These requirements propagate directly to the electrical power subsystem, where regulated DC rails must be delivered with high availability to heterogeneous loads (servomotors, sensors, microcontrollers, processing units), while preserving stability over rapidly changing operating points and minimizing the risk of electromagnetic interference with mission-critical electronics.

Within an avionics power-distribution unit, switching converters enable efficient step-down conversion from battery buses to regulated low-voltage rails. However, the same switching action that enables high efficiency and compactness also introduces coupled design pressures. On one side, pushing the switching frequency upward reduces passive size and improves transient performance, but moves the design into a regime where semiconductor switching losses, thermal management, and layout parasitics become dominant constraints. On the other side, the pulsed input-current spectrum must comply with stringent conducted-emission limits, typically verified through LISN-based measurements and standardized masks. In space-oriented avionics, ECSS electromagnetic-compatibility constraints are therefore a key driver: the input filter cannot be treated as a late add-on, because it can interact with the converter input dynamics and alter the effective plant seen by the controller. For this reason, the filter design must be carried out such that the converter dynamics are preserved according to the extra-element interaction criteria (e.g., Middlebrook’s extra element theorem), while still providing the required attenuation at the switching fundamental.

Developed in collaboration with Sidereus Space Dynamics during an industrial internship, this thesis targets a concrete and demanding operating point: a high-power synchronous buck converter intended to generate an avionics rail from a 10S Li-ion battery bus. Due to harness drops, the effective input range at the converter terminals spans approximately $V_{in} \in [26.93, 41.95]$ V, while the regulated output is specified as $V_o = 13.5$ V ($\pm 5\%$) to guarantee about 12 V at remote load terminals. The current envelope is high (32 A nominal, up to 50 A peak in worst-case conditions), with simultaneous constraints on efficiency (greater than 90%), output ripple, and an ECSS conducted-emission limit of 96 dB μ V. These combined requirements favor a high switching frequency but also increase sensitivity to non-idealities and implementation de-

lays across both the power stage and the control loop.

From the control perspective, fully digital regulation is attractive because it enables reconfigurable and repeatable control laws and supports traceable validation from model to hardware. Gains and setpoints can be tuned without redesigning analog networks, and practical features such as reference shaping (soft-start/soft-stop) and saturation with anti-windup can be implemented deterministically. At the same time, high switching frequencies typically motivate high sampling and update rates, which turn control computation into a strict real-time budget. In a microcontroller, the algorithm must execute sequentially within each sampling period and its latency can become comparable to, or larger than, the available margin. An FPGA is therefore preferred in high-speed regimes, because the controller is realized as parallel hardware rather than instruction code, enabling deterministic, cycle-bounded latency and eliminating firmware-induced jitter and overhead. Even with FPGA-based control, performance is constrained by ADC latency, fixed-point quantization, and DPWM delay and resolution. In particular, conventional counter-based DPWMs require exponentially higher clock rates as resolution increases; for this reason, delta-sigma noise-shaping techniques are a well-established approach to increase effective duty-cycle resolution without pushing the DPWM clock into impractical ranges [19, 24, 12].

The literature on digitally controlled DC-DC converters and FPGA-based implementations is broad. FPGA controllers have been proposed in reconfigurable and application-driven forms [22, 4, 20], and fixed-point FPGA realizations of voltage-mode control have been reported [3]. Other works emphasize the digital modulator and resolution bottleneck, including quantization-noise shaping and multi-bit $\Sigma\Delta$ PWM strategies [19, 24]. Fully digital alternatives to classical fixed-frequency PWM regulation, such as constant-on-time approaches, are also widely explored [13, 16]. However, many contributions focus primarily on one axis (controller structure, modulation resolution, or digital architecture) and only partially address the end-to-end integration problem under realistic high-power constraints: power-stage sizing under simultaneous efficiency and thermal limits, standards-driven conducted-emission compliance through an input filter designed to preserve plant dynamics, and a delay-aware digital controller that is not only functionally correct but also timing-closed and verifiable on a constrained FPGA device. The objective of this thesis is to address this gap by treating power stage, EMC filtering, control design, quantization, timing closure, and verification as a single engineering chain.

To this end, the thesis delivers the following contributions:

- A complete high-power synchronous buck converter design derived from avionics-level requirements, including component sizing, loss budgeting, and thermal verification under worst-case operating conditions.

- An input EMI filter designed after the power stage and tuned to meet the ECSS conducted-emission limit at the selected switching frequency, while preventing significant interaction with the converter dynamics according to extra-element stability criteria.
- A delay-aware digital control design synthesized from the modeled plant: the compensator is designed in continuous time and mapped to a discrete-time PID compensator while explicitly accounting for ADC conversion, computation, and DPWM update effects; quantization and DPWM limitations are treated as primary constraints.
- A fully custom FPGA implementation of the complete acquisition-to-actuation chain in fixed-point arithmetic, including soft-start, saturation with anti-windup, a feasible-resolution DPWM, and an error-feedback $\Delta\Sigma$ stage to increase effective duty-cycle resolution without adding delay on the signal path.
- A multi-level verification flow combining system simulations (MATLAB and PSIM), behavioral and post-implementation timing verification, and hardware-in-the-loop experiments using an external microcontroller-based ADC emulator to provide high-speed, reproducible stimulus.

The final outcome is a co-designed solution that meets the key application constraints simultaneously: EMC compliance through a quantitatively sized and properly damped input filter, a delay-aware digital control loop suitable for high-frequency operation, and a timing-closed FPGA implementation that preserves deterministic behavior while leaving headroom for monitoring features and multi-rail expansion.

This thesis is organized as follows: **Chapter 1** introduces the aerospace/avionics context and motivates the use of FPGAs in flight-oriented electronic systems; **Chapter 2** reviews the state of the art of FPGA-based digital control for DC–DC converters and highlights the practical bottlenecks (latency and DPWM resolution); **Chapter 3** presents the synchronous buck design, including specifications, loss/thermal sizing, and EMI-driven input filter design; **Chapters 4–5** develop the discrete-time control framework and derive the delay-aware digital PID (including quantization/resolution aspects); **Chapter 6** details the FPGA implementation and the verification flow; **Chapter 7** discusses results and positions the work against the reference literature, and the final chapter concludes the thesis and outlines future work.

3.2 Specifications and Design Requirements

The Buck converter designed in this work has been conceived as part of the **Avionics Power Distribution Unit** of a launch vehicle. Its purpose is to provide a regulated **12 V power rail** for several on-board subsystems such as servomotors, sensors, microcontrollers, and processing units. The converter receives its input from the **on-board battery pack**, while supplying multiple loads distributed along different points of the vehicle.

A summary of the key electrical specifications and design requirements is provided in Table 3.1.

Parameter	Symbol	Nominal / Limit Value	Notes
Input voltage range	V_{in}	27–42 V (battery) / 26.93–41.95 V (effective)	10S Li-ion battery pack
Output voltage	V_{out}	13.5 V $\pm$ 5%	Ensures $\sim$ 12 V at the load terminals
Nominal output current	$I_{out,nom}$	32 A	Sum of all load currents
Maximum output current	$I_{out,max}$	50 A	25% of servos in stall condition
Efficiency @ nominal load	η_{nom}	$\geq 90\%$	Design optimization target
Efficiency @ full load	η_{FL}	$\geq 80\%$	For $I_{out} = I_{out,max}$
Conducted EMI limit	$I_{line,noise}$	≤ 96 dB μ V	According to ECSS Avionic standard
Voltage ripple	ΔV_{pp}	$\leq 1\% \times V_{out,nom}$	—
Switching frequency	f_s	To be optimized	Trade-off between size and losses

Table 3.1: Key electrical specifications and limits

3.2.1 System Context

The input source consists of a **10S Li-ion battery pack**, whose voltage varies between 42 V (fully charged) and 27 V (fully discharged), with a nominal voltage of **36 V**. However, the converter is not directly connected to the battery terminals: an input harness—comprising cables and DC connectors—introduces a **voltage drop** due to its finite resistance.

3.3 Power Stage Design

3.3.1 Selection of Power MOSFETs

The MOSFETs are the key switching elements in the buck converter, directly influencing its efficiency, switching frequency, and thermal behavior. Their selection must therefore balance **conduction losses**, **switching losses**, and **thermal constraints**, all while meeting voltage and current stress requirements under nominal and transient conditions.

Power Loss Analysis in MOSFETs

The selection and optimization of the power MOSFETs can be rigorously performed by analyzing the **switching behavior** of the devices through their **small-signal equivalent model**.

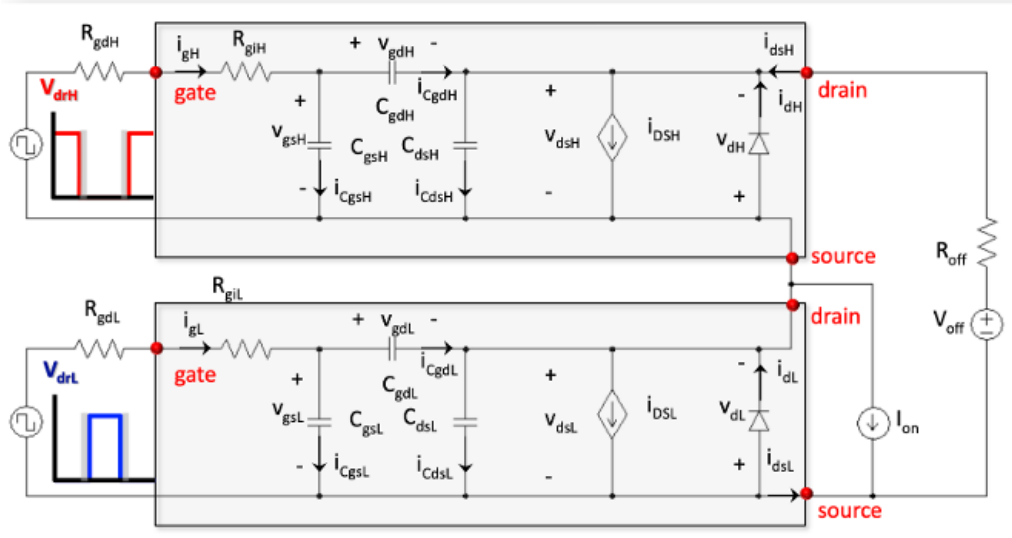


Figure 3.3: MOSFETs switching model

This model, shown in Figure 3.3, represents each transistor (high-side and low-side) including its **non-ideal parasitic elements**, the intrinsic capacitances C_{gs} , C_{gd} and C_{ds} , the gate resistance R_g and the body diode characterized by a forward voltage drop V_F and reverse-recovery charge Q_{rr} .

In a **synchronous buck converter**, the two MOSFETs operate complementarily:

- The **high-side** (HS) MOSFET connects the input voltage V_{in} to the inductor during the on-time interval DT_s ;
- The **low-side** (LS) MOSFET provides a current path to ground during the off-time $(1 - D)T_s$.

During each switching period, the parasitic capacitances of both devices interact with the driver circuit and with each other, defining the **dynamic**

By iteratively solving the steady-state thermal equilibrium equation:

$$T_j = T_{amb} + R_{\theta JA} P_d(T_j) \quad (3.31)$$

and using a measured effective *junction-to-ambient thermal resistance* of $2.94\text{ }^{\circ}\text{C/W}$, the following results were obtained at 730 kHz switching frequency and are summarized in Table 3.3.

Condition	I_{out} [A]	$T_{j,HS}$ [$^{\circ}\text{C}$]	$P_{d,HS}$ [W]	$T_{j,LS}$ [$^{\circ}\text{C}$]	$P_{d,LS}$ [W]
Nominal	32	87.0	19.19	44.0	4.6
Maximum	50	132.4	34.79	57.1	9.38

Table 3.3: Thermal performance of the high-side (HS) and low-side (LS) MOSFETs under nominal and maximum load conditions.

At maximum load, the high-side (HS) MOSFET exhibits a junction temperature that exceeds the maximum target value (set to 60% of its rated limit), reaching $T_{j,HS} = 132.9^{\circ}\text{C}$, while still remaining below the absolute rating of 150°C . Conversely, the low-side (LS) MOSFET remains well within its thermal margin, with $T_{j,LS} = 57.1^{\circ}\text{C}$.

The dissipated power of both devices exceeds their respective allocated budgets ($P_{d,HS,max} = 19.2\text{ W}$, $P_{d,LS,max} = 4.8\text{ W}$), which is expected under maximum current conditions, as the converter efficiency decreases due to increased conduction and switching losses.

Overall, the thermal results confirm safe operation for both devices. The HS MOSFET operates close to its thermal limit but remains within its rated temperature, while the LS MOSFET shows a wide safety margin. The power budget exceedance is consistent with the expected reduction in efficiency at full load and does not compromise device reliability.

Thermal Management and Layout Considerations

To ensure efficient heat dissipation, both MOSFETs exploit the exposed thermal pad of the PowerPAK[®] SO-8 package. Heat flows from the silicon junction to the ambient through the following chain:

$$R_{\theta JA} = R_{\theta JF} + R_{\theta FB} + R_{\theta BS} + R_{\theta TIM} + R_{\theta SA,ch}. \quad (3.32)$$

The contributions used in this design are:

- $R_{\theta JF} = 1.25\text{ }^{\circ}\text{C/W}$: junction-to-foot resistance of the PowerPAK SO-8 (from the MOSFET datasheet).
- $R_{\theta FB} = 0.20\text{ }^{\circ}\text{C/W}$: foot-to-board transition (solder layer and dense via-in-pad network).

3.3.2 Output Filter Design (L, C, ripple)

The output filter plays a crucial role in ensuring voltage stability and current continuity at the converter output. Its purpose is twofold: to attenuate the high-frequency switching components of the PWM signal and to maintain the output voltage ripple and transient deviation within the design specifications.

In this work, the filter was designed following **analytical criteria** that take into account both steady-state and dynamic constraints and its parameters were later refined through simulation and component characterization.

The design process begins with the establishment of performance specifications. For the buck converter, the nominal output voltage was set to $V_{out,nom} = 13.5 \text{ V}$, with a $\pm 5\%$ **regulation window**, supplying a nominal load current of $I_{out,nom} = 32 \text{ A}$ and a maximum load current of $I_{out,max} = 50 \text{ A}$. The converter operates at a switching frequency of $f_s = 730 \text{ kHz}$ and an efficiency of approximately $\eta = 0.9$.

A **voltage ripple** not exceeding 1% of $V_{out,nom}$ and a dynamic deviation within $\pm 5\%$ during **load transient** were imposed design constraints. The relationships governing the inductor current ripple and the resulting output voltage ripple are shown in Figure 3.6, where the steady-state operation of the buck converter is illustrated.

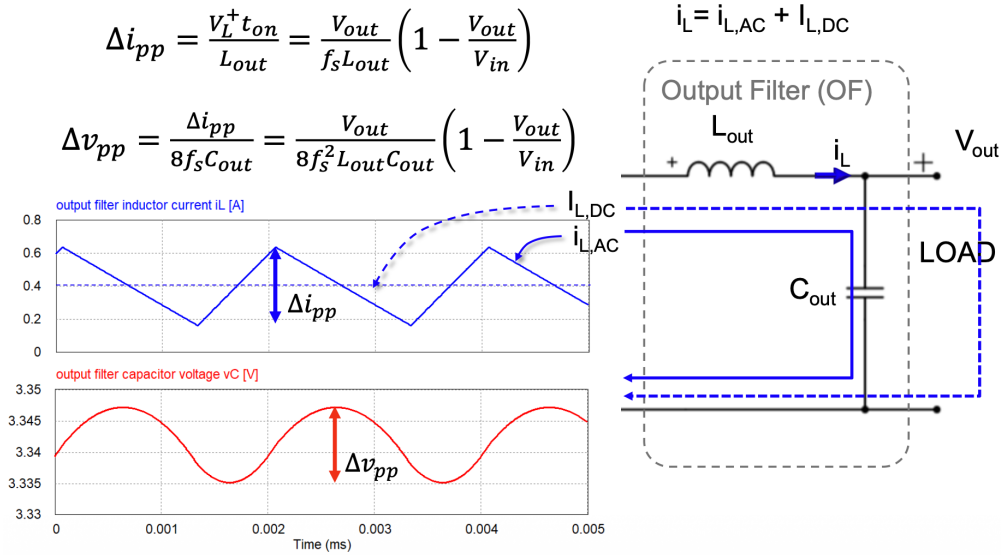


Figure 3.6: Inductor current and output capacitor voltage ripple in the Buck converter output filter

The top plot depicts the inductor current i_L , composed of a DC component $i_{L,DC}$ and an AC ripple component $i_{L,AC}$, with a peak-to-peak variation Δi_{pp} determined by the converter duty cycle and switching frequency. The lower plot shows the corresponding output voltage ripple Δv_{pp} across the output capacitor C_{out} , which is the combined response of the converter to the inductor current ripple.

During a sudden load step, the capacitor must provide or absorb the difference between the load current and the inductor current. The transient voltage deviation ΔV_0 is expressed as:

$$\Delta V_0 = \frac{\Delta I_{out}}{C} \Delta t + \Delta I_{out} \text{ ESR} \quad (3.36)$$

where the first term represents the capacitive effect, and the second accounts for the instantaneous voltage drop across the ESR. By combining this expression with the converter's regulation window, it is possible to derive analytical limits for the minimum capacitance and the maximum ESR required to maintain the output voltage within the specified $\pm 5\%$ range during a transient event. The reference expressions are:

$$C_{ref,lt} = C_o \frac{(1 - D_{nom})}{r} \left(1 + \frac{r}{2}\right)^2 \cdot 1.1, \quad ESR_{ref,lt} = \frac{R_o}{\left(1 + \frac{r}{2}\right)} \cdot 0.9 \quad (3.37)$$

where

$$r = \frac{\Delta I_{Lpp}}{\Delta I_o}, \quad R_o = \frac{(\Delta V_{reg} - \Delta V_{acc})}{\Delta I_o}, \quad C_o = \frac{1}{f_s R_o}.$$

The parameter r defines the ratio between inductor current ripple and load current variation, while R_o and C_o represent, respectively, the equivalent dynamic resistance and capacitance of the converter at the switching frequency. A 10% safety margin was again included to account for manufacturing and thermal variations.

These relationships highlight that, for fast load steps, the capacitive term dominates the transient behavior, making the load-transient constraint typically more restrictive than the ripple one. For this reason, the value $C_{ref,lt}$ obtained from this condition generally defines the **minimum effective capacitance** required for voltage recovery and thus it was adopted as the reference value in the design.

Comparing both constraints, the **load-transient requirement** resulted to be more demanding for capacitance, while the **ripple constraint** dictated the maximum admissible ESR. The final design parameters were therefore:

$$C_{ref} = 68.8 \mu F, \quad ESR_{ref} = 7.38 m\Omega \quad (3.38)$$

corresponding respectively to the load-transient and ripple specifications.

Once the reference values C_{ref} and ESR_{ref} were established from the ripple and load-transient constraints, the **number of the capacitor** to be placed in parallel was determined to ensure compliance with all electrical and thermal limits. For each practical capacitor, three independent conditions must

A single-stage **LC input filter** is adopted; its intent is to force the pulsed current to recirculate locally through C_{in} , leaving the line current nearly DC. The practical filter must also be **stabilized** (damped) to avoid peaking near its natural resonance and to respect the converter's control stability (Middlebrook extra-element perspective).

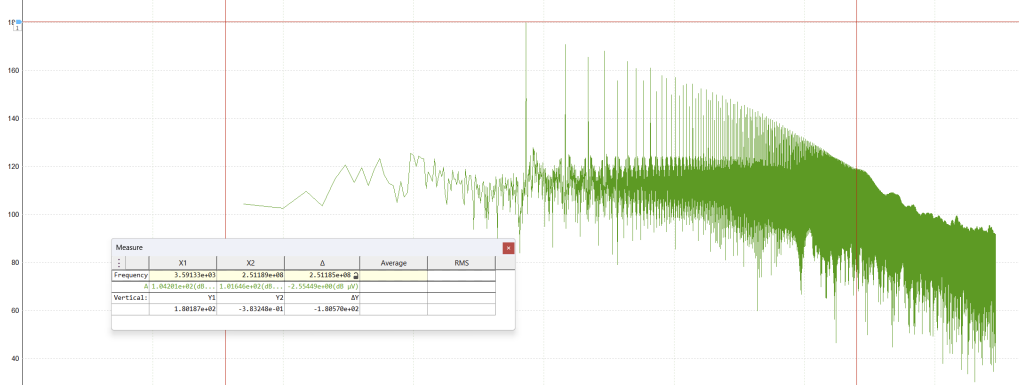


Figure 3.8: Line current harmonic level at switching frequency f_s , measured at 180 dBμV without input filter.

Input capacitor sizing

The input capacitor is dimensioned from line ripple and current stress constraints. Imposing a 2% peak-to-peak ripple on the input voltage,

$$\Delta V_{in,pp,max} = 0.02 V_{in} \quad (3.54)$$

the ESR and capacitance minima follow (with a 10% safety margin as in the script):

$$ESR_{max} = 0.9 \frac{\Delta V_{in,pp,max}}{I_{L,pk}}, \quad C_{min} = 1.1 \frac{D(1-D)I_{out}}{f_s \Delta V_{in,pp,max}} \quad (3.55)$$

The capacitor RMS current is estimated as

$$I_{C_{in},rms} = I_{out} \sqrt{D(1-D) + \frac{1}{12} \left(\frac{V_{out}}{I_{out} f_s I_{out}} \right)^2 + (1-D)^2 D} \quad (3.56)$$

Numerical results:

$$\begin{aligned} I_{L,pk} &= 41.51 \text{ A}, & ESR_{max} &= 15.6 \text{ m}\Omega, \\ C_{min} &= 16.28 \text{ }\mu\text{F}, & I_{C_{in},rms} &= 21.37 \text{ A} \end{aligned} \quad (3.57)$$

A **Würth 870056175005** electrolytic (33 μF, 10.7 mΩ, 2.8 A rms) is selected and paralleled in $N = 8$ to satisfy all constraints:

$$C_{in} = 264 \text{ }\mu\text{F}, \quad ESR_{C_{in}} = 1.3375 \text{ m}\Omega \quad (3.58)$$

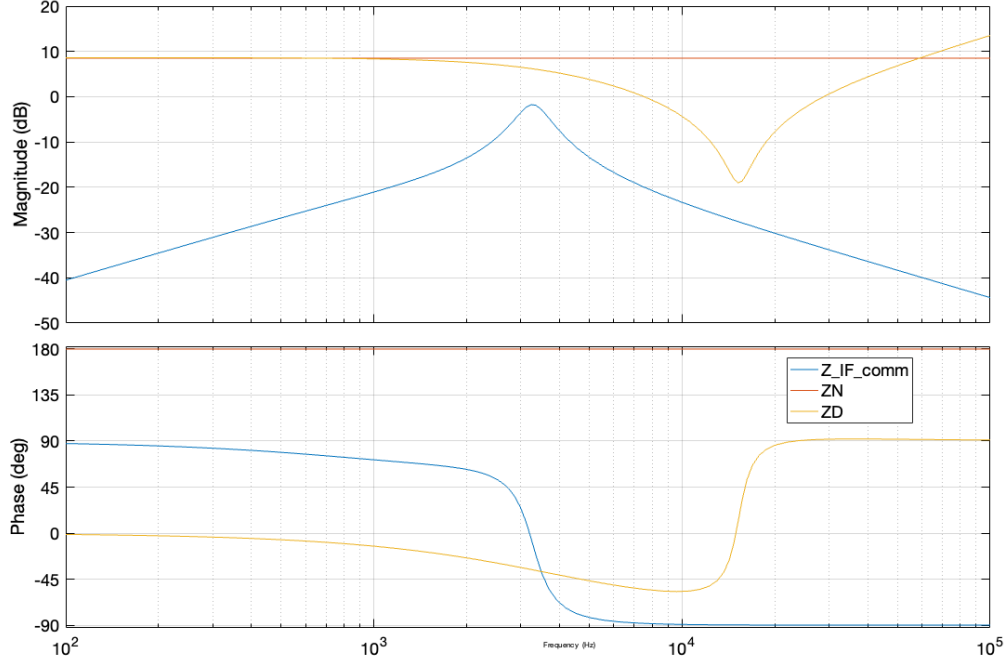


Figure 3.10: Bode plot of $Z_{out,IF}$ compared to Z_N and Z_D . The damping network attenuates the resonance, ensuring stability and EMI compliance.

Attenuation at f_s and EMI compliance

The complete (parasitic-aware) attenuation from line to the switching node is:

$$Att_{comm}(s) = \frac{ESRC_{in} + \frac{1}{sC_{in}}}{ESRC_{in} + \frac{1}{sC_{in}} + [(R_d + sL_d)^{-1} + (R_f + sL_f)^{-1}]^{-1}}. \quad (3.82)$$

At $f_s = 730$ kHz,

$$Att_{comm}(f_s) \approx 87.56 \text{ dB}, \quad (3.83)$$

$$V_{noise,filtered} = 180 - 87.56 \approx 92.44 \text{ dB}\mu\text{V} < 96 \text{ dB}\mu\text{V}.$$

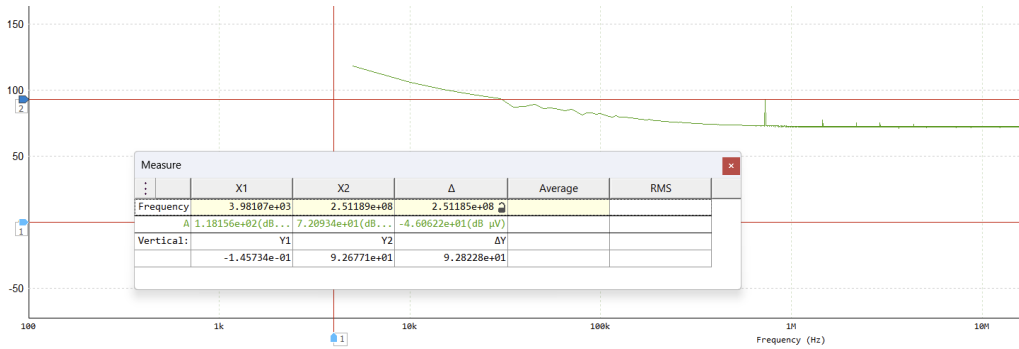


Figure 3.11: Line current noise spectrum attenuated by the input filter, showing a fundamental level of approximately 92 dBμV at $f_s = 730$ kHz.

Chapter 5

Digital PID Control

This chapter develops, realizes, and validates a digital PID controller for the synchronous buck converter operated in Voltage-Mode Control. Starting from the plant models and specifications established in Chapter 3, the objective is to achieve a **crossover frequency** $f_c \approx f_s/10$ with adequate phase margin while accounting for non-idealities specific to digital control. The compensator is first designed in continuous time as a PI with one or two lead networks (depending on the required phase boost), then mapped to the discrete domain.

The design explicitly includes the timing budget of the digital loop, A/D acquisition and conversion, computational latency, and DPWM action, which is modeled as a sample delay in the loop gain. The analog anti-alias pole is retained in the sensing path rather than in the controller. Discretization uses the bilinear (Tustin) transform with pre-warping at ω_c for the compensator, and a zero-order hold for the plant path, yielding a sampled-data loop $L(z)$ suitable for frequency-domain verification.

Implementation aspects are then addressed: coefficient extraction, difference-equation forms amenable to FPGA deployment, anti-windup and duty-cycle saturation, and the impact of quantization in the ADC and DPWM, including conditions to avoid limit cycles. Finally, the controller is verified in PSIM under line/load transients and worst-case operating points.

5.1 Digital PID Design

The digital PID controller is derived from the **analog compensator** previously designed in Chapter 3 for the buck converter operating in Voltage Mode Control. The analog controller, implemented as **PI+lead** network, provided a crossover frequency $f_c = f_s/10 = 73 \text{ kHz}$ and a phase margin of approximately 52°. In the transition to the digital domain, the objective is to preserve the same closed-loop dynamics while accounting for sampling, computational delays, and A/D conversion effects.

5.1.1 Analog-to-Digital Conversion (ADC)

For this application, a **Successive Approximation Register (SAR)** ADC was selected, offering an excellent trade-off between speed, resolution, and

5.1.4 Continuous-to-Discrete Mapping

Starting from the analog compensator derived in Chapter 3, the digital implementation is obtained by discretizing the continuous-time transfer function $G_c(s)$ through the bilinear (Tustin) transform with pre-warping at the crossover frequency ω_c . This ensures that both the magnitude and phase of the digital compensator match those of the analog design at f_c .

The analog compensator with **single lead** is expressed as:

$$G_c(s) = G_{cm} \underbrace{\left(1 + \frac{\omega_L}{s}\right)}_{\text{PI}} \cdot \underbrace{\frac{1 + s/\omega_z}{1 + s/\omega_{p1}}}_{\text{lead (PD)}}, \quad (5.15)$$

while the **double-lead** version, adopted when a larger phase boost is required, becomes:

$$G_c(s) = G_{cm} \underbrace{\left(1 + \frac{\omega_L}{s}\right)}_{\text{PI}} \cdot \underbrace{\left(\frac{1 + s/\omega_z}{1 + s/\omega_{p1}}\right)^2}_{\text{doppio lead (PD)}} \quad (5.16)$$

Both compensators are discretized using MATLAB's `c2d` function with pre-warping at the crossover frequency:

$$G_c(z) = \text{c2d}(G_c, T_{ctrl}, 'prewarp', \omega_c), \quad (5.17)$$

which applies the substitution

$$s = \frac{2}{T_{ctrl}} \cdot \frac{1 - z^{-1}}{1 + z^{-1}} \cdot \frac{\omega_c T_{ctrl}/2}{\tan(\omega_c T_{ctrl}/2)}. \quad (5.18)$$

After discretization, the compensator is represented in **zero–pole–gain form**. For the **single-lead** case, the resulting discrete transfer function is:

$$G_c(z) = G_d \frac{(z - z_L)(z - z_Z)}{(z - 1)(z - z_p)}, \quad (5.19)$$

with:

- a pole at $z = 1$ (integrative action),
- a second pole z_p (discrete image of ω_{p1}),
- and two zeros z_L, z_Z (integrator and lead zeros).

For the **double-lead** case, the discrete form becomes:

$$G_c(z) = G_d \frac{(z - z_L)(z - z_Z)^2}{(z - 1)(z - z_p)^2}, \quad (5.20)$$

corresponding to two identical lead sections that provide a distributed phase boost and a smoother high-frequency roll-off.

To compensate for the limited hardware resolution of the DPWM, a **second-order error-feedback $\Delta\Sigma$ modulator** (Fig.5.5) is introduced between the output of the digital compensator and the DPWM input. This choice does not introduce additional poles in the forward path (no extra delay on the useful signal) and performs **noise shaping** of the quantization noise toward high frequencies, where the power stage, being low-pass, naturally filters it out.

We denote:

- $v_{ch}[n]$: high-resolution PID output (here UQ1.15).
- $a[n]$: quantizer input (sum of useful signal and filtered feedback).
- $v_c[n]$: quantized output with n_{DPWM} bits, then mapped back to the normalized domain for computations.
- Subtractive error:

$$e[n] \triangleq a[n] - v_c[n] \quad \Rightarrow \quad v_c[n] = a[n] - e[n]. \quad (5.50)$$

The feedback filter processes only the error and reinjects it at the input node:

$$f[n] = (g * e)[n] \quad \Longleftrightarrow \quad F(z) = \mathcal{Z}\{g[n]\}. \quad (5.51)$$

Hence, the input sum is

$$a[n] = v_{ch}[n] + f[n]. \quad (5.52)$$

Derivation in the z -domain.

The two basic relations are

$$A(z) = V_{ch}(z) + F(z)E(z), \quad V_c(z) = A(z) - E(z). \quad (5.53)$$

Eliminating $A(z)$ gives

$$V_c(z) = V_{ch}(z) + (F(z) - 1)E(z). \quad (5.54)$$

From this we read directly:

- **Signal Transfer Function (STF)** from the useful signal

$$\text{STF}(z) = \left. \frac{V_c}{V_{ch}} \right|_{E=0} = 1, \quad (5.55)$$

i.e., the PID command passes cleanly with no extra delay.

- **Noise Transfer Function (NTF)** from the quantization error

$$\text{NTF}(z) = \left. \frac{V_c}{E} \right|_{V_{ch}=0} = F(z) - 1. \quad (5.56)$$

To obtain classical second-order shaping, we choose, in the simplest form,

$$F(z) = -2z^{-1} + z^{-2} \quad (5.57)$$

$$\text{NTF}(z) = F(z) - 1 = -(1 - 2z^{-1} + z^{-2}) = -(1 - z^{-1})^2. \quad (5.58)$$

The leading minus sign is irrelevant in magnitude; in amplitude

$$|\text{NTF}(e^{j\omega})| = |1 - e^{-j\omega}|^2 \approx \omega^2 \quad (\omega \rightarrow 0), \quad (5.59)$$

i.e., low-frequency noise attenuation with a slope of approximately 40 dB/dec (in power).

Time-domain equations (discrete implementation).

From the chosen $F(z)$ it follows that

$$f[n] = -2e[n-1] + e[n-2], \quad (5.60)$$

and the modulator reduces to three operational lines (plus the state update of $e[n-1], e[n-2]$):

$$\begin{cases} a[n] = v_{ch}[n] + f[n], \\ v_c[n] = Q_{n_{\text{DPWM}}}(a[n]) \\ e[n] = a[n] - v_c[n]. \end{cases}$$

Here $Q_{n_{\text{DPWM}}}(\cdot)$ denotes the quantizer that retains only the most significant n_{DPWM} bits.

In DC ($z = 1$), $\text{NTF}(1) = 0$ (double zero): quantization noise does not pass in the base-band. With $\text{STF} = 1$, the PID signal reaches the output without dynamic degradation.

In the discrete domain, the shaping concentrates the noise toward **Nyquist** (high digital frequencies). After the DPWM this appears as a small duty **jitter** between periods; spectrally it clusters around f_s and its harmonics. The power stage, being low-pass, suppresses it: what appears on v_o is much smaller than without the $\Delta\Sigma$. Practically, the modulator **ditherizes** the duty among a few adjacent codes: the *average* value follows the PID, while the instantaneous grain is pushed out of band.

As a result, the $\Delta\Sigma$ modulator enables a **significant increase in effective resolution**, typically by 6 to 7 bits, without any increase in clock frequency. In the design, the 9-bit hardware DPWM therefore achieves an **effective temporal resolution of about 15 bits**, satisfying the condition $G_{d0}H_0q_{\text{DPWM}} < q_{A/D}$ and ensuring stable steady-state operation even with finite quantization.

5.3 Simulations and Validation in PSIM

The complete digital control architecture for the synchronous buck converter has been implemented in **PSIM**, as illustrated in Fig. 5.6. The schematic reproduces the entire **closed-loop system** at the *value level*, that is, using analog behavioral blocks to represent the digital operations of sampling, computation, and PWM modulation. This choice allows for the verification of dynamic behavior prior to the bit-level implementation on FPGA.

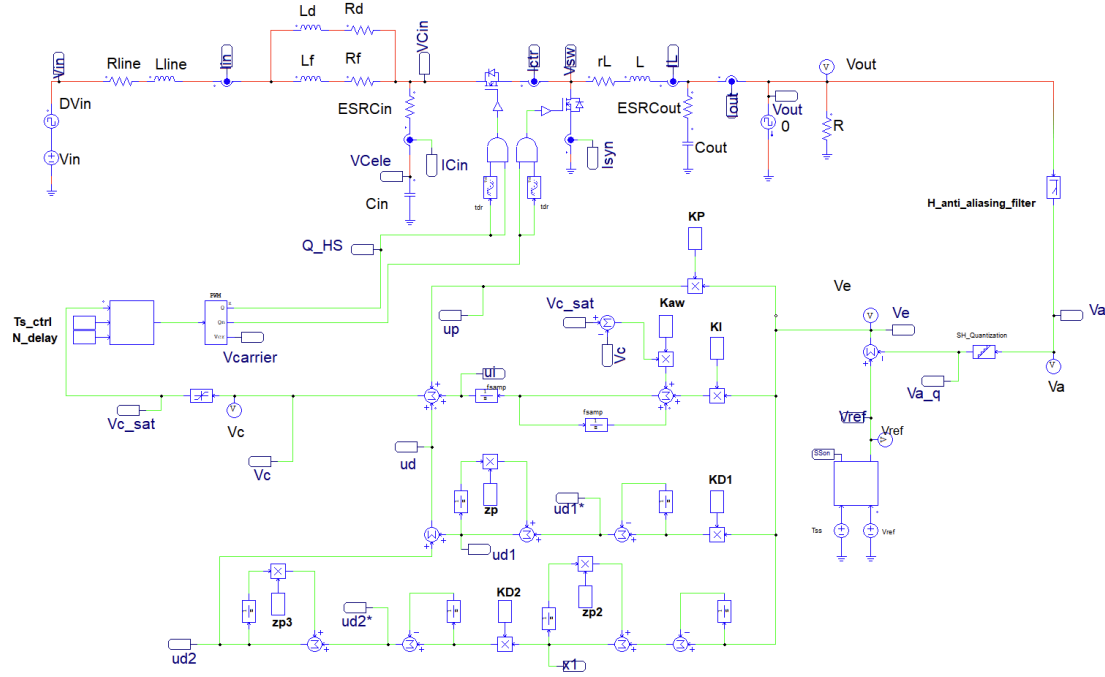


Figure 5.6: PSIM implementation of the digitally controlled buck converter.

At the top of the schematic, the **power stage** (plant) includes the input line model, input filter, high-side and low-side MOSFETs, inductor, capacitor and load resistance, faithfully representing the converter's switching dynamics and parasitics. The **feedback path** measures the output voltage V_{out} , which is passed through the **anti-aliasing filter**.

Following the filter, the **A/D conversion block** models the behavior of the 12-bit SAR ADC (Texas Instruments ADS7883) described earlier. The output of this block corresponds to the quantized voltage sample V_a , used as the input to the digital compensator.

The **digital compensator** implements the discrete-time version of the PI+lead (or double lead) controller obtained through the bilinear transform. Its structure is realized using **elementary arithmetic blocks** (adders, multipliers and delays) mirroring the discrete difference equation derived in Section 5.2.1. This modular approach allows the internal coefficients (K_P , K_I , K_D) and zeros/poles (z_L , z_Z , z_P) to be directly visualized and adjusted, facilitating verification of each control term prior to FPGA synthesis.

5.3.1 Load Transient Response

The first validation scenario concerns the **load transient response**, which evaluates the converter's ability to maintain voltage regulation when the load current abruptly changes.

Two opposite transitions were simulated in PSIM:

1. **Nominal-to-maximum load (L–H transition)**
2. **Maximum-to-nominal load (H–L transition)**

Both tests were carried out under nominal input voltage and reference conditions, with the digital controller operating at a sampling frequency of $f_{samp} = 4f_s = 2.92$ MHz.

The maximum allowable output voltage deviation is defined as:

$$\Delta V_{o,max} = \Delta V_{out_{reg}} - \Delta V_{out_{acc}} - \frac{\Delta v_{opp}}{2} \quad (5.65)$$

where:

$$\Delta V_{out_{reg}} = 0.05 V_{out_{nom}}, \quad \Delta V_{out_{acc}} = 0.01 V_{out_{nom}}, \quad \Delta v_{opp} = \frac{\Delta i_{L,pp}}{8f_s C_{out}}.$$

These relationships establish the voltage regulation limits used as reference in the subsequent validation sections, ensuring that the converter meets both steady-state and transient performance specifications.

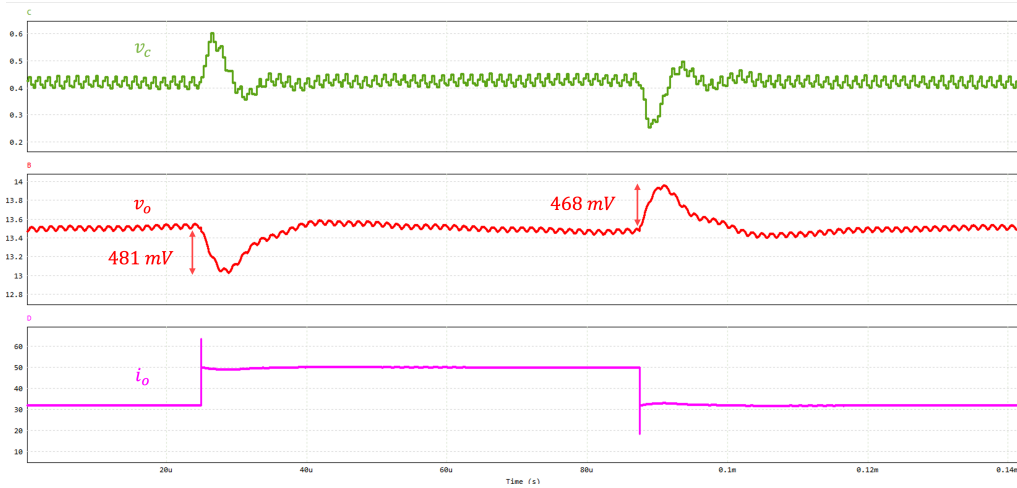


Figure 5.7: Load transient response in PSIM for nominal-to-maximum (L–H) and maximum-to-nominal (H–L) transitions

Nominal-to-Maximum Load (L–H Transition)

When the output current steps from its nominal value to the maximum load condition, the converter experiences an **undershoot** in the output voltage

6.2 Module-Level Design and RTL Coding

6.2.1 Overall architecture

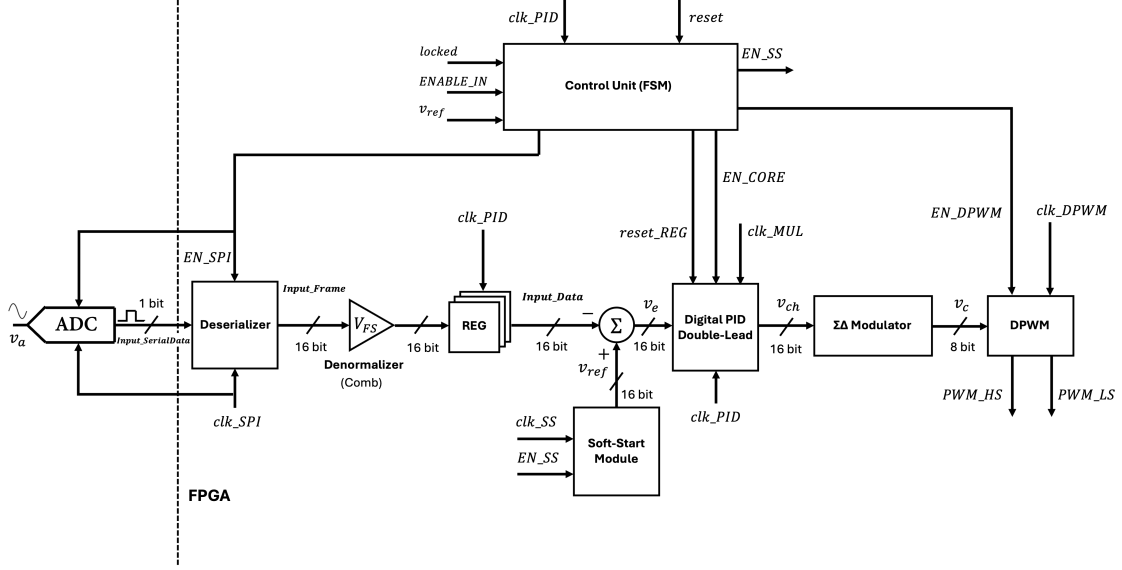


Figure 6.2: Digital Controller Block Diagram

The digital controller implemented on the FPGA realizes the complete ADC-to-DPWM chain shown in Fig. 6.2. The only external input is the output voltage of the synchronous buck converter, sampled by an external 12-bit SAR ADC (ADS7883) and delivered to the FPGA as a serial bitstream. Inside the FPGA, this stream is **deserialized** into a 16-bit word that mirror the SPI frame (00|D11...D0|00) and then transferred into the control domain through a register bank. The valid data bits are rescaled into a signed $Q4.12$ format, representing the sensed output voltage in per-unit form. A further denormalization step multiplies this value by the ADC full-scale ($V_{FS} = 5V$, encoded as $Q4.12$), so that the internal representation of the feedback signal coincides with the physical output voltage filtered by the analog anti-alias stage.

In parallel, a dedicated **soft-start module** generates the reference v_{ref} in the same $Q4.12$ format. Starting from zero, the reference ramps up to the target value $H_0 V_{out,nom}$ over a programmable number of switching periods, and can also be ramped down in a controlled way when the converter is disabled. To compensate for the **latency** introduced by the ADC interface and denormalization, v_{ref} is passed through a register stage clocked at the control frequency, ensuring temporal alignment between the reference and the measured output. The digital error v_e is then obtained as the difference between the delayed reference and the denormalized measurement and constitutes the input of the digital PID controller.

The **digital PID compensator** processes the error in discrete time and produces a high-resolution control word proportional to the desired duty cy-

rest of the digital controller operates on parallel words, this serial stream must be converted into a 16-bit vector that is stable and time-aligned with the discrete-time operation of the PID.

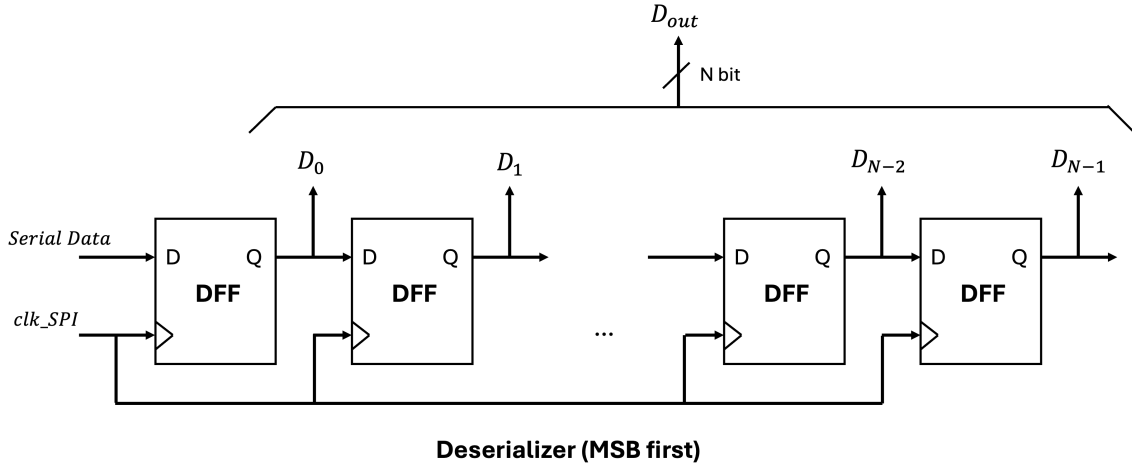


Figure 6.11: Deserializer (MSB first)

This task is carried out by the **Deserializer** module, implemented as a simple shift register of $N = 16$ flip-flops. All stages are clocked by clk_SPI and enabled by the signal EN_SPI . The incoming bit D_in is injected into the first stage, while at each rising edge of clk_SPI the content of every flip-flop is shifted towards the most significant position.

After exactly 16 clk_SPI cycles, which fit inside one sampling period of clk_PID , the whole frame corresponding to the current ADC conversion has been loaded into the register bank, and the output vector D_out exposes in parallel the last 16 bits received.

From the point of view of the controller, this introduces an effective **latency of one sampling period**: during interval k , the ADC acquires sample k and streams its 16-bit frame; by the end of the same interval the deserializer has completed the parallelisation, so that at the next rising edge of clk_PID (beginning of interval $k + 1$) the corresponding 16-bit word is already stable and can be safely used by the downstream blocks. In other words, one sample period is spent to serialize, transfer and deserialize the frame, but a new fully aligned conversion is still available at every control tick.

In the following stage, the 16-bit word provided by the deserializer is post-processed by discarding the two leading and two trailing zeros and extracting the 12-bit payload. These 12 bits are then zero-extended on the most significant side to obtain a 16-bit signed value in $Q4.12$ fixed-point format, which represents the normalized ADC output and is used as input to the denormalization multiplier and to the subsequent error computation feeding the digital PID control.

the complete closed loop shows that the converter output tracks this digitally generated ramp with negligible error, thanks to a crossover frequency set to about one tenth of the switching frequency.

Overall, the `Soft_Start_Module` provides a compact and fully programmable way to shape the reference trajectory, both at start-up and shut-down, directly in the FPGA, improving robustness and repeatability compared to purely analog implementations.

Digital PID Controller with Double-Lead Compensation

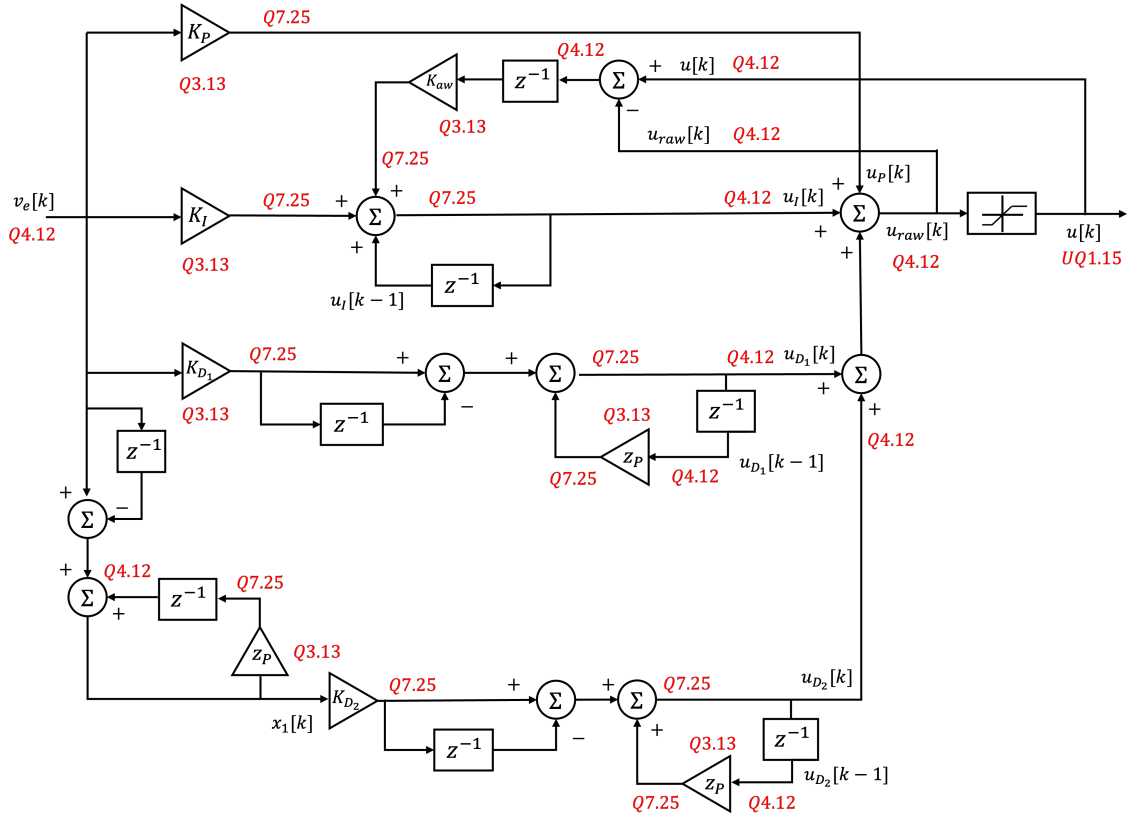


Figure 6.12: Annotated PID Architecture with Fixed-Point Formats

The module `Digital_PID_DoubleLead` implements the discrete-time PID controller with double-lead compensation derived in Chapter 5. It operates on the error signal $v_e[k]$ provided by the subtractor, represented in signed fixed-point format Q4.12 on 16 bits. All controller gains K_p , K_i , K_{aw} , K_{d1} , K_{d2} and z_p are pre-computed offline and stored as Q3.13 constants, so that every multiplication between a gain and the error (or its processed versions) produces a 32-bit result in Q7.25.

The proportional branch is conceptually the simplest: at each sample the controller computes

$$u_p[k] = K_p v_e[k]. \quad (6.28)$$

In hardware this is realized by an iterative Booth multiplier (BM1) clocked by `clk_MUL`, followed by a `Round_Shifter` that converts the product from

compute the quantization error $e[k]$ and update the internal error states $e_1[k]$ and $e_2[k]$, realizing the target noise transfer function:

$$\text{NTF}(z) = (1 - z^{-1})^2. \quad (6.37)$$

In this way, the modulator provides an 8-bit DPWM-ready duty code with shaped quantization noise, fully aligned with the timing and resolution constraints imposed by the power stage and the FPGA implementation.

DPWM module

The `DPWM_module` implements the final digital-to-time conversion stage, generating the complementary gate signals for the high-side and low-side MOSFETs from the quantized duty code $v_c[k]$ provided by the $\Delta\Sigma$ modulator. The block is fully parameterized in terms of DPWM resolution N_{DPWM} and dead-time, but in the final implementation $N_{\text{DPWM}} = 8$ is adopted to keep the DPWM clock frequency within the timing capabilities of the target FPGA while still providing a sufficiently fine duty resolution of $1/2^{N_{\text{DPWM}}}$ per switching period.

The core of the module is a free-running N_{DPWM} -bit counter `cnt` clocked by `clk_DPWM`. One DPWM period corresponds to $2^{N_{\text{DPWM}}}$ clock cycles, and at the end of each period (`cnt == {N_DPWM{1'b1}}`) the current duty code v_c is latched into `v_c_latched`. This ensures that the duty command is sampled only once per period and remains constant for the whole duration of the next switching interval, even if the upstream digital controller updates v_c at the faster control rate. The raw PWM waveform `PWM_RAW` is then generated by a simple unsigned comparison:

$$\text{PWM_RAW} = (\text{cnt} < v_{c,\text{latched}}), \quad (6.38)$$

which encodes the desired duty ratio as the fraction of the period during which `PWM_RAW` is high.

To safely drive a synchronous buck converter, complementary gate signals with guaranteed non-overlap are required. For this purpose, the module includes a small Moore FSM with five states (`IDLE`, `HS_ON`, `LS_ON`, `DEAD_TO_HS`, `DEAD_TO_LS`) and an internal dead-time counter.

The required dead-time in clock cycles is computed at compile time from the desired dead-time in nanoseconds and the DPWM clock frequency:

$$\text{DEAD_CYCLES} \approx \text{round}\left(\frac{\text{DEADTIME_NS} \cdot F_{\text{CLK_HZ}}}{10^9}\right), \quad (6.39)$$

and stored in the parameter `DEAD_CYCLES`.

During normal operation, each time `PWM_RAW` requests a transition from high-side to low-side conduction (or vice versa), the FSM first moves to the corresponding dead-time state (`DEAD_TO_LS` or `DEAD_TO_HS`) and keeps both outputs `PWM_HS` and `PWM_LS` low while `dt_counter` increments up to `DEAD_CYCLES`

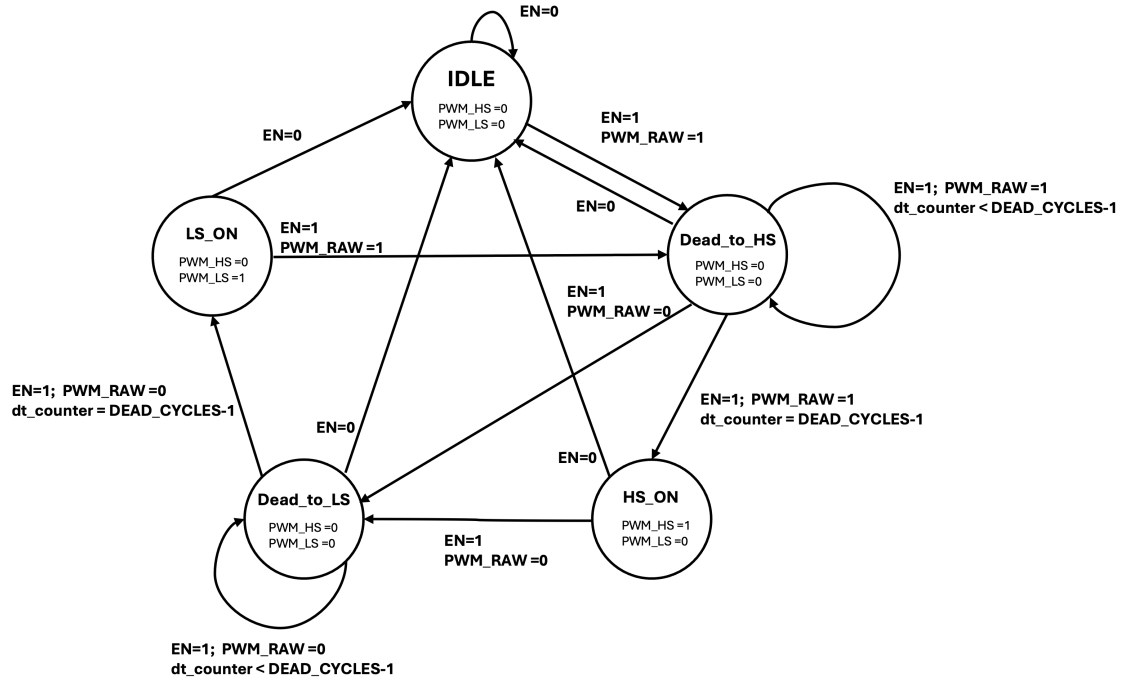


Figure 6.13: DPWM FSM used to enforce complementary gate driving with guaranteed dead-time.

- 1. Only after the programmed dead-time has elapsed does the FSM enable the new conduction state (HS_ON or LS_ON). To guarantee **glitch-free** gate commands, the FSM generates the high-side and low-side signals combinationally but updates them only through synchronous registers on the rising edge of `clk_DPWM`, ensuring that PWM_HS and PWM_LS can change solely at clock boundaries.

As a result, the generated PWM_HS and PWM_LS signals are both glitch-free and strictly non-overlapping for any duty code, the effective dead-time is directly specified in physical units (nanoseconds), and the resulting gate signals are ready to be fed to the external driver stage of the power converter.

ADC2DPWM_VMC_Top module

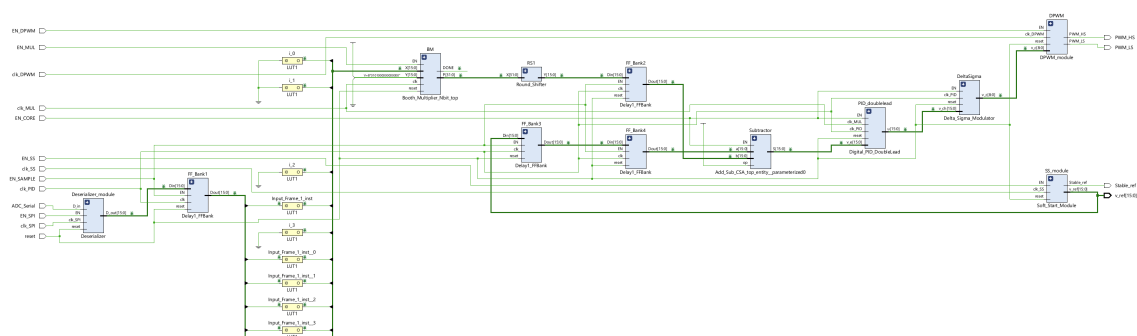


Figure 6.14: ADC2DPWM_VMC_TOP Schematic

The ADC2DPWM_VMC_Top module ties together all the previously described blocks, implementing the complete digital chain from the ADC serial out-

With the final 8-bit configuration and the constraints introduced in the previous sections, synthesis and implementation complete without timing violations. The post-implementation Timing Summary reports a worst setup slack (WNS) of +0.347 ns, a worst hold slack of +0.096 ns, and a minimum pulse-width margin of +0.522 ns, with zero failing endpoints in all categories. All user-defined timing constraints are therefore met.

6.4.6 Resource utilization, power and placement

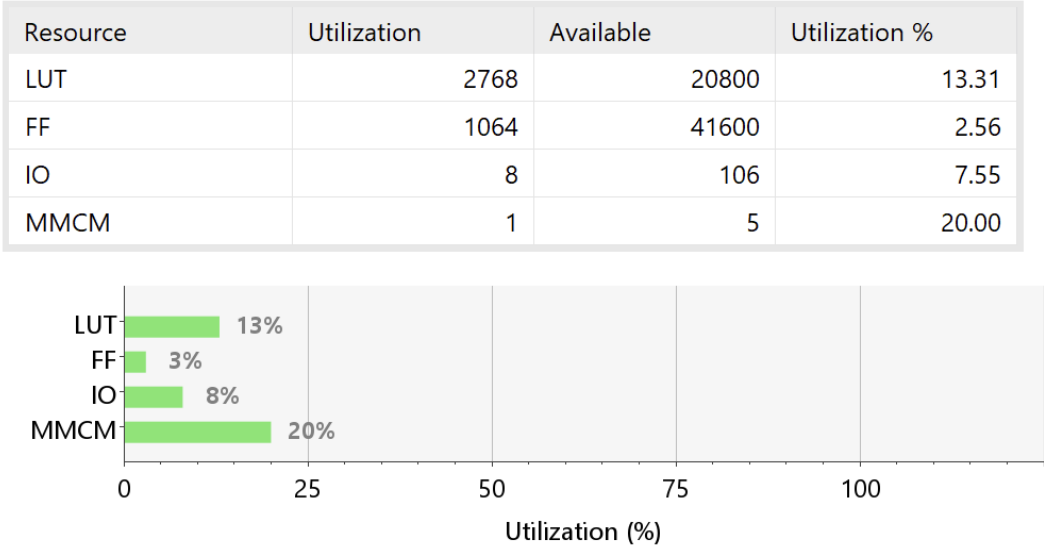


Figure 6.27: Utilization Report

Summary

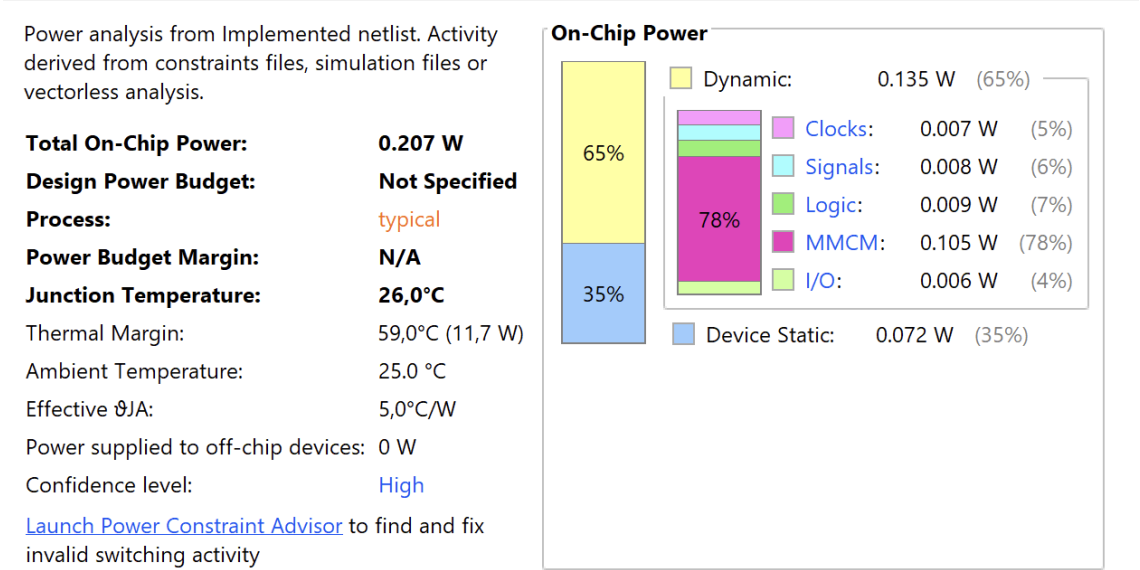


Figure 6.28: Power Report

The post-implementation **utilization report** shows that the controller occupies only a modest fraction of the XC7A35T resources. In particular, the

design uses 2768 LUTs (13.3% of the 20 800 available), 1064 flip-flops (2.56% of 41 600), 8 user I/O pins (7.55% of 106) and a single MMCM out of the five available on the device. The implementation view confirms that the logic is placed compactly in the central clock region, without routing congestion or scattered hot-spots.

Power analysis performed on the implemented netlist estimates a total on-chip power of about 0.207 W. Roughly 65% of this figure is dynamic (0.135 W) and 35% is static (0.072 W). Within the dynamic component, the MMCM accounts for the largest share (around 78%), while clocks, logic, internal signals and I/Os contribute only a few percent each. For an ambient temperature of 25 °C the resulting junction temperature is approximately 26 °C, leaving a thermal margin of almost 60 °C. No power is drawn by off-chip devices in this analysis, since the external gate driver and ADC are not included in the FPGA power model.

Overall, the synthesis and implementation results confirm that the proposed multi-clock digital controller fits comfortably in the Cmod A7-35T platform, closes timing with positive margins at the required switching frequency, and exhibits very low silicon and power utilization, leaving ample headroom for future extensions or additional monitoring logic.

6.4.7 Post-Implementation Simulation

As a final verification step, the controller was simulated using the *implemented* netlist produced by Vivado after place-and-route. In this **timing simulation** the structural netlist of the XC7A35T device is back-annotated with the cell and interconnect delays extracted from implementation, so that all paths include the actual MMCM, clock-divider and routing delays. The same SystemVerilog testbench used for the behavioral simulations was reused unchanged: the testbench still reads the sequence of 16-bit ADC frames from the `frames.txt` file, streams them on the serial input pin synchronously with `clk_SPI`, drives `reset` and `ENABLE_IN` according to the power-up sequence, and records all internal debug signals of the ADC2DPWM chain. In this way, the implemented design is exercised with exactly the same input stimulus used for the behavioral model, which allows a one-to-one comparison of the resulting waveforms.

The post-implementation waveforms, shown in Fig. 6.29, confirm that the *functional behaviour* of the controller is preserved after synthesis and place-and-route. The pipelined reference `v_ref_pipelined_debug`, the error signal `v_e_debug`, the controller output `u_debug` and the DPWM input code `v_c_debug` exhibit the same evolution already observed in the pre-synthesis simulations: the soft-start ramp, the settling to the nominal duty ratio, and the perturbations associated with the load transients are all correctly reproduced.

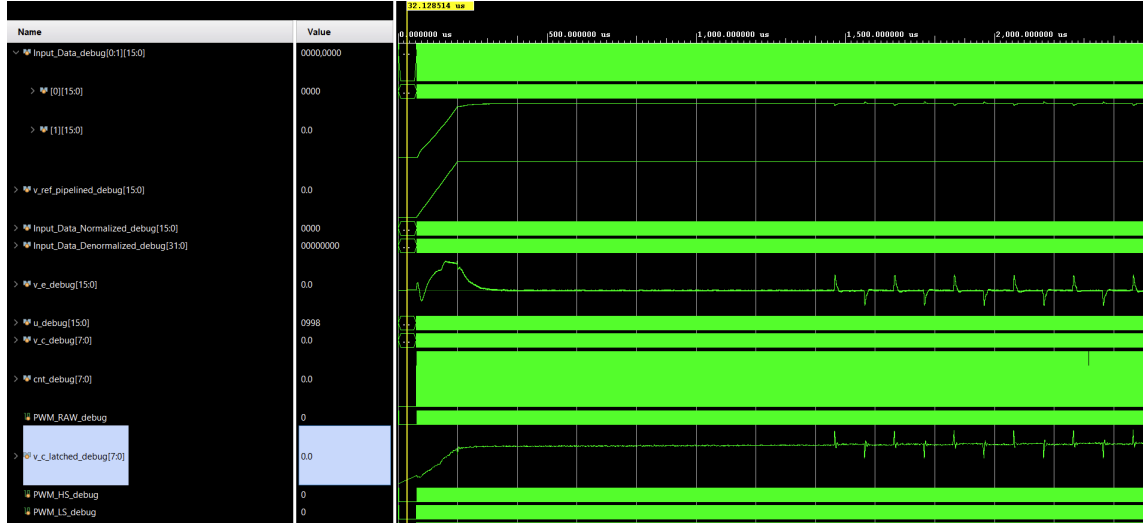


Figure 6.29: Post-Implementation simulation of the digital controller

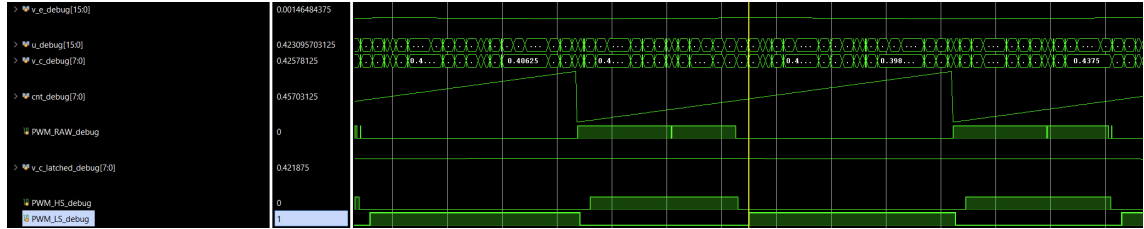


Figure 6.30: Post-implementation DPWM waveforms

The DPWM section also behaves as expected (Fig. 6.30): the raw PWM signal

PWM_RAW_debug, the latched duty command `v_c_latched_debug` and the gate signals PWM_HS_debug and PWM_LS_debug maintain the proper timing relationship dictated by the dead-time logic, with no spurious glitches or overlap between high-side and low-side pulses. The small additional skew visible on the signal edges is fully consistent with the propagation delays reported by the static timing analysis and remains well within the available slack. Overall, the timing simulation of the implemented netlist provides further evidence that the design meets the timing constraints on the target XC7A35T device while preserving the intended closed-loop behavior of the digital controller.

6.5 Hardware validation with STM32-based SPI ADC emulator

To validate that the implemented digital controller correctly receives the serialized ADC data stream, processes it and generates the expected PWM outputs also in hardware, a dedicated setup was assembled.

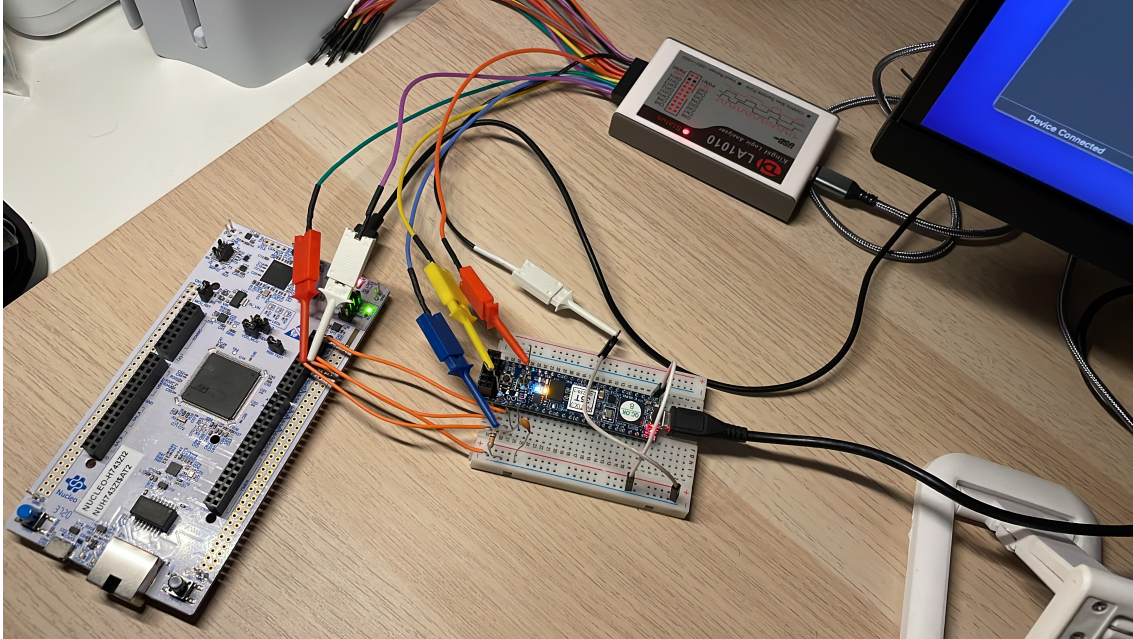


Figure 6.32: Experimental setup with the Cmod A7-35T FPGA and the STM32H743 ADC emulator

The FPGA implementation of the controller runs on the **Digilent Cmod A7-35T** development board (Artix-7 XC7A35T), which provides the external 12 MHz oscillator and the user I/O used throughout the project. Opposite to it, an STM32H743ZIT-based **Nucleo-H743ZI2** board is used as an “ADC emulator”.

The STM32 is configured as an SPI1 slave, with the SPI clock input SCK mapped to pin PA5 and the transmit line MISO mapped to PA6. In this configuration only the SCK and MISO lines are actually used: the STM32 behaves as a **transmit-only SPI slave** that outputs the ADC frames towards the FPGA, which plays the role of SPI master. A third digital line, connected to pin PB8 and labeled EN_SPI_IN in the firmware project, receives the enable signal coming from the controller’s control unit; this line is used by the microcontroller purely as a status indicator to detect when the FPGA has enabled or disabled the SPI interface.

The SPI timing is entirely dictated by the FPGA. Internally, the controller generates the SPI clock `clk_SPI` at approximately 46 MHz, which is then routed to the external pin `clk_SPI_out` through a tri-state buffer enabled by `EN_SPI_out`. When the finite-state machine enters the `PRIME_IO` and `RUN` states, `EN_SPI_out` is asserted, the buffer is activated, and the STM32 receives

a continuous stream of SCK edges on PA5. For each sampling period of the control loop, the controller generates exactly 16 clock pulses, allowing a full 16-bit frame to be shifted out from the SPI slave, fully mirroring the behavior of the actual ADC. As long as `EN_SPI_out` remains asserted, the SPI clock continues to run and the STM32 continuously acquires successive frames, each aligned to the rising edges of `clk_SPI_out`.

To ensure that this streaming can sustain the required clock frequency, the STM32 firmware does not rely on bit-banging of GPIOs, but uses the on-chip SPI1 peripheral together with DMA (Direct Memory Access), which is specifically designed to transfer data between memory and peripherals autonomously and at high throughput. The test dataset is exactly the same used in the behavioral simulations: starting from the PSIM waveform of the quantized ADC output `Va_q`, a MATLAB script converts the CSV file into 16-bit frames and automatically generates a C source file `frames_data.c` containing a `const uint16_t adcFrames[]` array and the corresponding frame count. At start-up, the firmware allocates a transmit buffer `spiTxBuf` with the same length as `adcFrames[]`, initializes it with the desired frame pattern, and then configures SPI1 in slave mode with 16-bit data size, `CPOL = 0`, `CPHA = 1st edge`, MSB first.

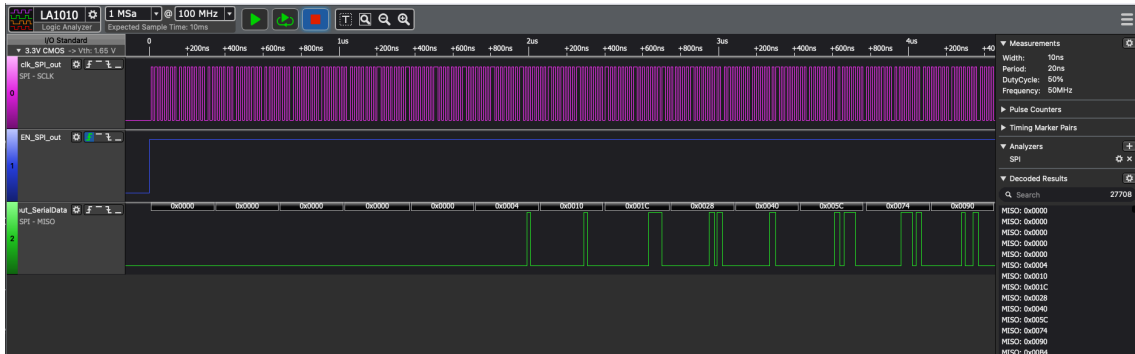


Figure 6.33: Logic Analyzer Capture of SPI Clock, Enable Signal, and Incoming Frames

The correctness of this hardware-in-the-loop test is verified using two complementary observation tools. On the one hand, an external USB logic analyzer is connected to `clk_SPI_out`, `EN_SPI_out` and to the serial data line from the STM32, and its decoded view confirms that each group of 16 clock pulses corresponds to the expected hexadecimal frame values generated from PSIM. On the other hand, an **Integrated Logic Analyzer (ILA)** core is instantiated inside the FPGA design and connected to the internal signals of the ADC2DPWM chain (deserializer output frame, decoded sample, reference, error signal and DPWM command). The simultaneous capture from the logic analyzer and the ILA shows that the serialized bitstream produced by the STM32 is sampled on the correct clock edges, that the reconstructed frames inside the FPGA exactly match the pre-computed test sequence, and that the internal fixed-point variables and PWM outputs follow the same trajectories observed in post-implementation simulation. This experiment there-

fore provides an additional level of confidence that the synthesized controller, when driven by a realistic high-speed SPI data stream, behaves exactly as designed.

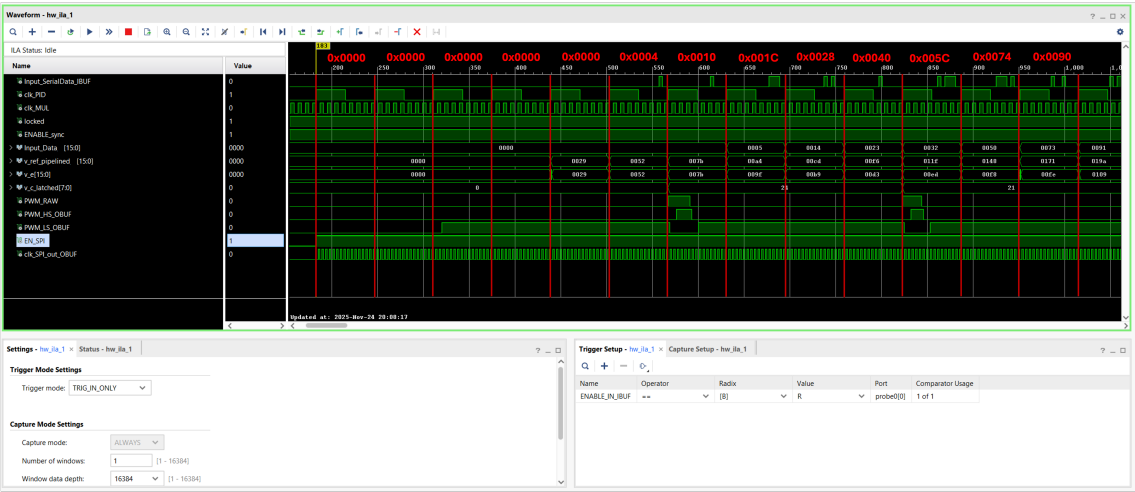


Figure 6.34: On-chip ILA acquisition of SPI frames and controller internal signals

Chapter 7

Results and Discussion

7.1 Power Stage and Digital Control Results

This section summarizes and discusses the results achieved across the entire design flow—power-stage modeling, digital control design, and FPGA implementation and prototyping—and illustrates how their integration validates the overall design approach of the proposed DC-DC converter for avionics power distribution.

The developed synchronous buck converter operates over the battery voltage range $V_{\text{BAT}} \in [27, 42]$ V—corresponding to an effective input range at the converter terminals of $V_{\text{in}} \in [26.93, 41.95]$ V once harness drops are accounted for—and provides a regulated 13.5 V rail (ensuring approximately 12 V at remote loads) under a demanding current envelope (32 A nominal, up to 50 A peak in worst-case transients).

The switching frequency is set to $f_s = 730$ kHz as a system-level compromise between passive component size, transient performance, switching-loss feasibility, and filterability of conducted emissions. The key design targets and corresponding verification results are summarized in Table 7.1.

The power stage was validated quantitatively through a detailed loss budget and a thermal assessment of the selected synchronous MOSFET pair. The analysis confirms compliance with device junction-temperature limits up to the maximum load condition, with the high-side MOSFET emerging as the dominant thermal bottleneck, as expected in a switching-loss-dominated regime. Moreover, the overall design satisfies the $> 90\%$ efficiency target, consistent with the intended avionics specifications and the adopted switching-frequency operating point.

Filtering choices were verified against both dynamic performance and EMC constraints. The output network was sized to satisfy steady-state ripple requirements while preserving load-transient capability, resulting in an LC resonance placed around ~ 20 kHz. At $f_s = 730$ kHz, the output filter provides 61.77 dB of attenuation, effectively suppressing switching harmonics without excessively penalizing control bandwidth. On the input side, the EMI filter was designed after the power stage and then sized to be essentially non-invasive for the converter dynamics: by satisfying the Middlebrook extra-element criterion, the plant transfer functions remain practically unchanged

at $f_{CPU} = 400$ MHz (and $\approx 5.50 \mu\text{s}$ in the higher-frequency configuration). Even in this optimistic scenario, the gap versus the required budget is:

$$\frac{T_{\text{calc,MCU}}}{T_{PID}} \approx \frac{5.36}{0.342} \approx 15.6 \approx 16. \quad (7.2)$$

This already exceeds the budget by about one order of magnitude. Consequently, the MCU-side timing implies a theoretical upper bound on the control update rate of $\approx 1/5.36 \mu\text{s} \approx 186$ kHz (core-only). In a realistic closed-loop implementation, once ADC sampling, PWM handling, interrupts, and control-software overhead are included, the achievable update rate is further reduced (typically into the tens of kHz, depending on the adopted peripheral strategy and firmware architecture).

This limitation is not merely a throughput inconvenience: it directly constrains the maximum feasible switching frequency for the intended scheduling. If the loop is designed such that the control computation must complete within each control tick (here explicitly set by $f_{PID} = 4f_s$), then any MCU-driven reduction of feasible f_{PID} forces a reduction of f_s . Lower f_s typically requires larger reactive components to meet ripple constraints, reduces achievable control bandwidth, and degrades load/line transient response, ultimately driving the implementation toward a slower and bulkier power stage. Conversely, the FPGA-based solution enables the targeted high- f_s , high- f_{PID} operating point while preserving deterministic timing and robust closed-loop behavior.

Parameter	Value	Conditions/Notes
Battery Voltage Range (V_{BAT})	27–42 V	Source voltage range
Input Voltage Range (effective)	26.93–41.95 V	Includes harness drop
Regulated Output Voltage	$13.5 \text{ V} \pm 5\%$	Ensures $\sim 12 \text{ V}$ at remote loads
Output Current Capability	32 A nominal, 50 A peak	Worst-case transient
Load transient performance	$\Delta V_{o,LH} \approx 0.481 \text{ V}$	Meeting the $\pm 5\%$ regulation specification
Efficiency Target	$> 90\%$	System-level requirement
Switching Frequency	$f_s = 730 \text{ kHz}$	Size/dynamics vs. losses/EMI trade-off
Output filter	$L \approx 675 \text{ nH}$, $C_{\text{out}} \approx 89.1 \mu\text{F}$, $ESR \approx 2.25 \text{ m}\Omega$	Output filter cutoff frequency $\sim 20 \text{ kHz}$
Conducted EMI @ f_s	$\sim 92.4 \text{ dB}\mu\text{V}$	Below 96 dB μV ECSS limit (with margin)
Control Sampling Frequency	$f_{\text{samp}} = 4f_s = 2.92 \text{ MHz}$	Control Update Rate
Total loop latency	$\sim 1.25 \mu\text{s}$	4 samples
DPWM base resolution/clock	$N_{\text{DPWM}} = 8$, $f_{\text{clk}} \approx 186.75 \text{ MHz}$	Timing Closure Confirmed (WNS +0.347 ns)
Resolution enhancement	2nd-order $\Delta\Sigma$	Improves effective duty resolution
Control crossover frequency/ PM	$f_c \approx 73 \text{ kHz}$, PM $\approx 52^\circ$	Stable closed-loop
FPGA resource utilization	LUT: 13.31%, FF: 2.56%, IO: 7.55%, MMCM: 20.00%	Post-implementation utilization report
Interface	Custom lightweight serial interface (SPI-like)	test-only streaming interface
On-chip power	Dynamic: 0.135 W, Static: 0.072 W	Implemented netlist power report (typical process, high confidence)

Table 7.1: Key power-stage, filtering, and digital-control results.

Conclusion and Future Work

This thesis has demonstrated the design and FPGA implementation of a high-power synchronous DC-DC buck converter for avionics power distribution, showing that deployability requires treating power-stage design, control design, timing, quantization, and implementation constraints as a unified co-design problem.

Developed with Sidereus Space Dynamics during an industrial internship, the work did not aim to introduce a new control technique in isolation; instead, its contribution lies in an integrated, end-to-end, industrially grounded methodology and architecture that remain consistent from specifications to implementation, with traceable requirements, realistic non-idealities, robustness-oriented design choices, and validation through both system-level simulations and implementation-level verification.

A key takeaway is that, in this class of applications, performance and reliability are dictated as much by real-time determinism and architectural suitability as by the compensator itself. By implementing the control loop as a custom RTL datapath on FPGA, the proposed approach delivers deterministic, cycle-bounded behavior under tight latency budgets imposed by sampling and modulation, while preserving loop bandwidth at high switching frequency.

At the same time, it enables the integration of the surrounding control infrastructure, soft-start, saturation, and anti-windup logic, the modulation/DPWM chain, without firmware-induced jitter or hidden overhead. Overall, the resulting end-to-end workflow and timing-closed implementation provide a solid baseline for hardware integration and further qualification steps in an aerospace-oriented development path.

Future work naturally follows the same industrial integration direction pursued throughout the thesis:

- **Custom PCB integration of power stage and FPGA controller:** design and manufacture a dedicated board hosting both the power converter and the FPGA, with an aerospace-oriented layout flow (power-plane partitioning, return-path control, high-current routing, decoupling strategy, and EMI-aware placement/routing). Stack-up definition and BGA fan-out for the FPGA will be key aspects to ensure signal integrity on high-speed clocks/interfaces while safely handling high di/dt power loops.
- **Multi-rail / multi-converter expansion on the same FPGA:** given the demonstrated implementation headroom on the target device, repli-

cate the controller chain to drive multiple regulated rails in parallel (e.g., the two main avionics rails) while adding monitoring, interlocks, and fault management logic.

- **Extension to alternative topologies:** port the same FPGA control framework to more advanced architectures relevant to avionics power distribution, such as interleaved/multiphase bucks (for reduced ripple and improved thermal sharing) and isolated topologies (e.g., flyback or forward-derived solutions) where galvanic isolation and wider input ranges are required.
- **Aerospace qualification alignment and SEU/bit-flip robustness:** evolve the design flow toward stronger compliance with aerospace standards and practices (including EMC-oriented verification) and increase resilience against radiation-induced upsets by adopting rad-tolerant/rad-hard FPGA solutions or radiation hardening by design techniques. This includes SEU/bit-flip mitigation in both configuration and user logic via periodic configuration scrubbing/readback, TMR (or selective TMR) on critical state machines and control datapaths, and error detection/correction mechanisms (e.g., parity/ECC on memories and safety monitors with fault logging and safe-state fallback).

Overall, the thesis demonstrates that high-frequency, high-update-rate digital control for a power converter can be realized as a complete, timing-closed, and verifiable engineering system on a constrained FPGA—in a way that is directly scalable toward flight-oriented integration and qualification.